Zhengyang Hu

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Profile

PhD student at HKU, Department of Electrical and Computer Engineering, working on **Dependence Modeling**, **Foundational Models for Data Science**, and **LLM Interpretability**. First-author publications at **ICML 2024 (Oral)** and **ICML 2026**; long-term collaborator with Huawei's HPC group (RAFT). Solid mathematical background and end-to-end research experience spanning theory, synthetic-data generation, and multi-GPU cluster pre-training.

Education

The University of Hong Kong — Dept. of Electrical and Computer Engineering, PhD Student 2023.08 – present
Advisors: Dr. Yanchao Yang & Prof. Kaibin Huang

University of Science and Technology of China (USTC) — School of the Gifted Young, B.Sc. 2019.09 – 2023.06
Major: Mathematics and Applied Mathematics • Major-Course GPA: **3.87 / 4.3**

Publications & Submissions

[1] InfoNet: Neural Estimation of Mutual Information without Test-time Optimization

Z. Hu, S. Kang, Q. Zeng, K. Huang, Y. Yang. **ICML 2024 (Oral)**.

- First to bring the foundation-model paradigm to mutual information estimation: a Transformer pre-trained on large-scale synthetic data zero-shot estimates the MI of arbitrary 1-D pairs (X, Y); $\sim$**100**$\times$ **faster** than prior neural methods, with even slightly better accuracy.

[2] A Foundation-style Model for Zero-Shot Statistical Dependency Measurement

Z. Hu, Y. Chen, H. Ren, Q. Zeng, Y. Zheng, A. Weller, K. Huang, Y. Yang. **ICML 2026**.

- Introduced **InfoAtlas**, a full upgrade of InfoNet: redesigned architecture and synthetic dataset supporting arbitrary dimensions; pre-trained $\sim$1 month on a 32$\times$H200 cluster into a 1B-parameter model, zero-shot ready and evaluated across broader downstream tasks.

[3] From Anisotropy to Anomaly: Online Geometric Diagnostics during Transformer Training

Z. Hu, W. Fang, Y. Zheng, Y. Yang. **Under review at NeurIPS** (collaboration with Huawei).

- An online Transformer-training monitor built on latent-space statistical correlation and geometric anisotropy; jointly characterizes geometric drift and representation anomalies, enabling real-time perturbation detection on NanoGPT / ViT / Pythia-2.8B.

Experience & Competitions

Lingjun Investment Quant Competition — A-Share Market Microstructure Prediction (Kaggle) 2026.03 – 2026.04

- Team ranked **Top 13–15**. Predicted intraday multi-horizon returns across 500 A-share stocks from Level-2 high-frequency features; benchmarked XGBoost / LightGBM / GRU / LSTM / Transformer with Optuna tuning; the single XGBoost reached **test R^2 = 0.030** (matching the runner-up's best single model, $R^2 \approx 0.029$); ensembling was skipped due to time.

Huawei HPC Group (RAFT) — LLM Training Dynamics & Interpretability, Visiting Researcher 2024.09 – 2026.01

- Designed real-time representation-monitoring metrics for training; evaluated across multi-scale Transformers and validated for early warning of training anomalies.

Chengqi Asset Management — Quantitative Strategy Research 2022.08 – 2022.12

- Built, backtested, and shipped 5 alpha factors that were integrated into the firm's factor / strategy library.

Languages : **TOEFL iBT** 105 (R 28 / L 25 / S 22 / W 30) • **GRE General** 322 (V 152 / Q 170 / AW 3.5)